

Monte Carlo Valuation of Contingent Coupon Callable on Three Stocks

1 Summary

In this project, we aim to value a Callable Contingent Income Security of three underlying assets via Monte Carlo simulation. Its underlying assets are the S&P 500 Index (SPX), Nasdaq-100 Technology Sector Index (NDXT) and Russell 2000 Index (RTY), and each modeled as a correlated geometric Brownian motion under the risk-neutral measurement.

Using a Least-Squares Monte Carlo (LSMC) to handle the call feature, the estimated value of this security is \$954.5.

2 Data

Interest rate, dividend yields, correlation between stocks and implied volatilities are downloaded from Bloomberg on the pricing date March 10th, 2025.

- **Interest Rate**

Using SWDF command in Bloomberg, we extract discount rate starting on March 10, 2025 (pricing date) from OIS term structure. Note that following market convention, the interest rates are quoted on March 8, 2025. Using linear interpolation, we get the discount rate through December 15, 2026 (maturity date) is 0.934719, the interest rate converted from this value is risk-free rate.

	Sep. 11, 2026	Dec. 15, 2026	Mar. 11, 2027
Discount rate	0.943391	0.934719	0.926869
Risk-free rate	3.86150%	3.82028%	3.80272%

Table 1: Discount rate of OIS

- **Dividend Yields**

Using OVME command in Bloomberg, we get dividend yield for three stocks between March 10, 2025 (pricing date) and December 15, 2026 (maturity date), which is actually calculated by method from OPDF command in Bloomberg.

$$\delta = (\delta_{SPX}, \delta_{NDXT}, \delta_{RTY}) = (1.439\%, 0.628\%, 1.548\%).$$

- **Correlation**

Using HRA command in Bloomberg, we calculate historical Pearson correlations between three stocks (in sequence of SPX, NDXT, RTY):

$$\rho = \begin{pmatrix} \rho_{SPX,SPX} & \rho_{SPX,NDXT} & \rho_{SPX,RTY} \\ \rho_{NDXT,SPX} & \rho_{NDXT,NDXT} & \rho_{NDXT,RTY} \\ \rho_{RTY,SPX} & \rho_{RTY,NDXT} & \rho_{RTY,RTY} \end{pmatrix}$$

According to Efficient Market Hypothesis, in a semi-strong efficient market, prices reflect all publicly available information as well as historical price information. So we can estimate future correlation by historical correlation, these relationships are already embedded in past data and tend to persist over time. Nonetheless, the correlations are not completely constant, a short term estimation window may reflect the impact of recent market sentiment and unexpected events, a medium term estimation window may capture some industry cycles and seasonal trends and a long term estimation window may be dominated by structural factors like some macroeconomic indicators. For simplicity, we assume correlations stay the same during our future estimation, hence it is important to choose the historical estimation period to appropriately match future correlations.

Here, we use weekly closing price data to estimate their correlations over short term (6 months), medium term (1 year) and long term (5 years):

$$\rho^{6m} = \begin{pmatrix} 1 & 0.861 & 0.910 \\ 0.861 & 1 & 0.767 \\ 0.910 & 0.767 & 1 \end{pmatrix} \quad \rho^{1y} = \begin{pmatrix} 1 & 0.895 & 0.767 \\ 0.895 & 1 & 0.620 \\ 0.767 & 0.620 & 1 \end{pmatrix} \quad \rho^{5y} = \begin{pmatrix} 1 & 0.868 & 0.858 \\ 0.868 & 1 & 0.741 \\ 0.858 & 0.741 & 1 \end{pmatrix}$$

• Implied Volatilities

Using OVM command in Bloomberg, we get volatility surface of stocks, we interpolate the potential volatility on the observation dates preceding the redemption dates and maturity date as follows:

Moneyiness	Jul. 10, 2025	Sep. 10, 2025	Dec. 10, 2025	Mar. 10, 2026	Jul. 10, 2026	Sep. 10, 2026	Dec. 10, 2026
60%	0.51517	0.42055	0.37504	0.34673	0.32646	0.31182	0.30078
70%	0.42183	0.35024	0.31769	0.29775	0.28419	0.27501	0.26937
100%	0.20387	0.19745	0.19478	0.19326	0.19375	0.19281	0.19350

Table 2: Volatility Matrix of SPX

Moneyiness	Jul. 10, 2025	Sep. 10, 2025	Dec. 10, 2025	Mar. 10, 2026	Jul. 10, 2026	Sep. 10, 2026	Dec. 10, 2026
60%	0.27612	0.27449	0.25533	0.2455	0.24379	0.24379	0.24379
70%	0.27612	0.27449	0.25533	0.2455	0.24379	0.24379	0.24379
100%	0.27612	0.27449	0.25533	0.2455	0.24379	0.24379	0.24379

Table 3: Volatility Matrix of NDXT

Moneyiness	Jul. 10, 2025	Sep. 10, 2025	Dec. 10, 2025	Mar. 10, 2026	Jul. 10, 2026	Sep. 10, 2026	Dec. 10, 2026
60%	0.49521	0.41406	0.39407	0.35803	0.35139	0.33552	0.32176
70%	0.4327	0.36669	0.34411	0.3156	0.31274	0.30107	0.2912
100%	0.26355	0.25098	0.24937	0.23867	0.24206	0.23764	0.23459

Table 4: Volatility Matrix of RTY

In our pricing process, we assume a constant volatility for each stock, but actually a single implied volatility cannot fully describe market expectations during different period. Therefore we will select the volatility corresponding to some key payment characteristics to test the price sensitivity to them.

Additionally, we notice that the volatilities of NDXT are the same for different moneyiness, and actually there is no market data for NDXT options, because there are no NDXT options being traded. With doubts about the validity of these data, future work could try volatility from other sources, such as historical realized volatility or implied volatility from other index options that are strongly correlated with NDXT.

Event	Date
Pricing Date	2025-03-10
Issue Date	2025-03-13
Final Observation	2026-12-10
Maturity Date	2026-12-15
Call Dates (t_τ)	2025-06-13, 2025-09-15, 2025-12-15, 2026-03-13, 2026-06-15, 2026-09-15
Observation Dates (t_m)	Monthly from 2025-04-10 to 2026-12-10
Coupon Payment Dates (t_m^{CPN})	Monthly from 2025-04-15 to 2026-12-15

Table 5: Key Dates for the Callable Contingent-Income Note

Parameter	SPX	NDXT	RTY
Initial Level S_0	5614.56	9548.55	2019.067
Downside Threshold (60%)	3368.736	5729.13	1211.440
Coupon Barrier (70%)	3930.192	6683.985	1413.347
Annual Coupon Rate		7%	
Notional Principal		\$1 000	

Table 6: Key Prices for Underlying Stocks

3 Algorithm

We simulate N correlated paths of the three underlying $S_{SPX}, S_{NDXT}, S_{RTY}$ on the grid of M observation dates using risk-neutral geometric Brownian motion. Denote by $\mathbf{S}_{n,m} \in \mathbb{R}^3$ the vector of index levels for path n at time step m , and let $\Delta t_m = (t_{m+1} - t_m)/365$. We perform:

$$\mathbf{S}_{n,m+1} = \mathbf{S}_{n,m} \exp\left((r - \boldsymbol{\delta} - \frac{1}{2}\boldsymbol{\sigma}^2) \Delta t_m + \boldsymbol{\sigma} L \Delta \mathbf{W}_{n,m}\right),$$

where:

- r is the continuously compounded risk-free rate.
- $\boldsymbol{\delta}$ is the vector of continuously compounded dividend rate.
- $\boldsymbol{\sigma}$ is the vector of volatility $(\sigma_{SPX}, \sigma_{NDXT}, \sigma_{RTY})$.
- L is the lower-triangular Cholesky factor of the correlation matrix.
- $\Delta \mathbf{W}_{n,m} \sim \mathcal{N}(\mathbf{0}, \Delta t_m I_3)$ are independent standard normals.

3.1 Contingent Coupon

A contingent coupon at an annual rate of 7.00% will be paid on the securities on each coupon payment date but only if the closing level of each underlier is greater than or equal to its coupon barrier level on the related observation date:

$$CPN_{n,m} = P \cdot c \cdot \frac{t_m - t_{m-1}}{365} \cdot \mathbf{1}_{\{S_{n,m,j} \geq \text{Coupon Barrier}_j, \forall j\}}$$

where:

- $CPN_{n,m}$ denotes the contingent coupon paid on coupon date t_m^{CPN} for n -th path.
- P is the notional principal.

- c is the annual coupon rate 7%.
- t_m is the m -th observation date.

3.2 Payment at Maturity

At maturity, each path n receives

$$\text{Payment}_{n,M} = CPN_{n,M} + P \times \underbrace{\begin{cases} 1, & \text{if } \forall j, S_{n,M,j} \geq \text{Downside Threshold}_j, \\ \min_j \frac{S_{n,M,j}}{S_{0,j}}, & \text{otherwise} \end{cases}}_{\text{principal repayment}}.$$

Then on last observation date, the discounted value is:

$$CV_{n,M} = \text{Payment}_{n,M} \cdot e^{-r(t_M^{CPN} - t_M)}.$$

3.3 Payment on observation dates

On observation dates, we will discount the payment from later observation dates and contingent coupon from following coupon payment dates. That is,

$$CV_{n,m} = CV_{n,m+1} \cdot e^{-r(t_{m+1} - t_m)} + CPN_{n,m} \cdot e^{-r(t_m^{CPN} - t_m)}$$

3.4 Least-Squares Monte Carlo (LSMC) for Call Feature

We work backwards through the set of redemption dates $\{t_r\}$. At each call date:

1. For $t_m^{CPN} \in \{t_r\}$, compute discounted continuation value $CV_{n,m}$ from later cash flows.
2. Form orthogonal polynomials $\{B_k(S)\}$ on stock prices $S_{n,m,i}$ with order k .
3. Estimate continuation by least-squares regression:

$$\hat{CV}_{n,m} = \sum_i \sum_k \theta_{k,i} B_k(S_i), \quad \boldsymbol{\theta} = (S^\top S)^{-1} S^\top \mathbf{CV},$$

where $X_{n,k} = B_k(x_{n,j})$.

4. Compare $\hat{CV}_{n,m}$ to the immediate call value:

$$V_{n,m}^{\text{call}} = (P + CPN_{n,m}) \cdot e^{-r(t_m^{CPN} - t_m)}$$

Then,

$$CV_{n,m} = \begin{cases} V_{n,m}^{\text{call}}, & \text{if } V_{n,m}^{\text{call}} < \hat{CV}_{n,m}, \\ \hat{CV}_{n,m}, & \text{else} \end{cases}.$$

3.5 Algorithm Summary

1. Generate N paths via `simulate_stock_price`.
2. Compute terminal payoffs with `payment_at_maturity`.
3. Backward induction with `Callable_Contingent_pricer` discounting security value and contingent coupon at each observation date, as well as performing LSMC at each call date.
4. Discount all cash flows to pricing date and average over paths:

$$\hat{V} = \frac{1}{N} \sum_{n=1}^N CV_{n,1} \cdot e^{-r(t_1 - t_{\text{pricing date}})}$$

4 Discussion

4.1 Numerical Sensitivity

We assess the impact of LSMC regression basis and Monte Carlo sample size N on this security's value.

Regression Basis In LSMC, we use orthogonal polynomials of different orders and forms as regression independent variables. By our test, replacing Legendre with Chebyshev polynomial basis in the LSMC regression alters the price by less than 0.001%, confirming negligible impact from basis selection. Additionally, the order of polynomials also shows little influence on estimation. So we then choose three order Legendre polynomials for LSMC.

Number of Simulation Paths By law of large numbers, as simulation paths increase, the average of simulation values should converge to the real mean and standard deviation should decrease. In figure 1, we vary the simulations from 10,000 to 500,000, as simulation increases, the standard deviation reduces, the confidence interval becomes narrower and the estimated value is convergent. We then choose 500,000 simulations for Monte Carlo.

Overall, the impact of numerical settings on estimated value is not as great as that of the financial data itself. So then we will discuss the sensitivity of financial parameters.

4.2 Financial Sensitivity

Correlation Window The correlation matrix calculated for different time periods has a clear impact on the price. Since we assume the stock prices follow geometric Brownian motions, and when the correlations are different, the normal random values obtained by our simulations will be different as well, resulting in different simulated prices. As this security has a maturity of 21 months, we similarly use a medium-term historical correlation matrix to estimate future medium-term correlations. The figure 2 shows estimated prices of 100,000 simulations under different volatilities, where downside and barrier level are 60% and 70% of initial stock prices as described in the file, respectively.

Volatility Specification Simulated stock prices vary if volatilities in geometric Brownian motions are different. Generally, if the volatility is too small, the probability of hitting coupon barrier will be low, the security seldom pays coupon, the probability of being called decreases, resulting in an undervaluation, and if the volatility is too big, the probability of hitting coupon barrier will be too high, the security pays coupon too often, the probability of being called increases, resulting in an undervaluation.

Here we choose volatilities corresponding to some special payment level and analyze their influences.

(1) ATM Implied Volatility $K = S_0$: Captures near-the-money uncertainty; highly liquid but underestimates large barrier-crossing probabilities.

(2) Coupon Barrier-Level Implied Volatility $K = 0.70 S_0$: Directly reflects market pricing of touches at the 70% coupon-call barrier; yields more accurate trigger probabilities.

(3) Downside Threshold-Level Implied Volatility $K = 0.60 S_0$: Quantifies extreme downside risk; critical for estimating principal-at-risk losses if the worst-of underlier breaches the 60% threshold.

We have chosen to use the *barrier-level implied volatility* on last observation date as our single volatility input for two main, academically grounded reasons:

(1) Payoff-Relevance. The product's coupon payments and early-call features hinge directly on whether the underlying stays above (or first crosses) the 70% barrier. Barrier-level IV captures the market's view of price uncertainty precisely at that strike, yielding a more accurate estimate of the relevant event probabilities than ATM volatility.

(2) Model-Payoff Consistency. It measures the overall volatility of the stocks over the life of the security. The barrier-level volatility thus serves as a coherent single-factor proxy that is both theoretically justified and practically implementable.

The figure 3 shows estimated price of 500,000 simulations under different volatilities and correlations.

Conclusion: The estimation is sensitive to the volatilities and correlations among three underlying stocks, and essentially invariant to the form of regression. The final estimated value for the Callable Contingent Income Securities is \$954.5 based on our method and selected parameters (500,000 simulations with 3-order Legendre polynomials basis in LSMC, 1 year correlation matrix and volatility at barrier level).

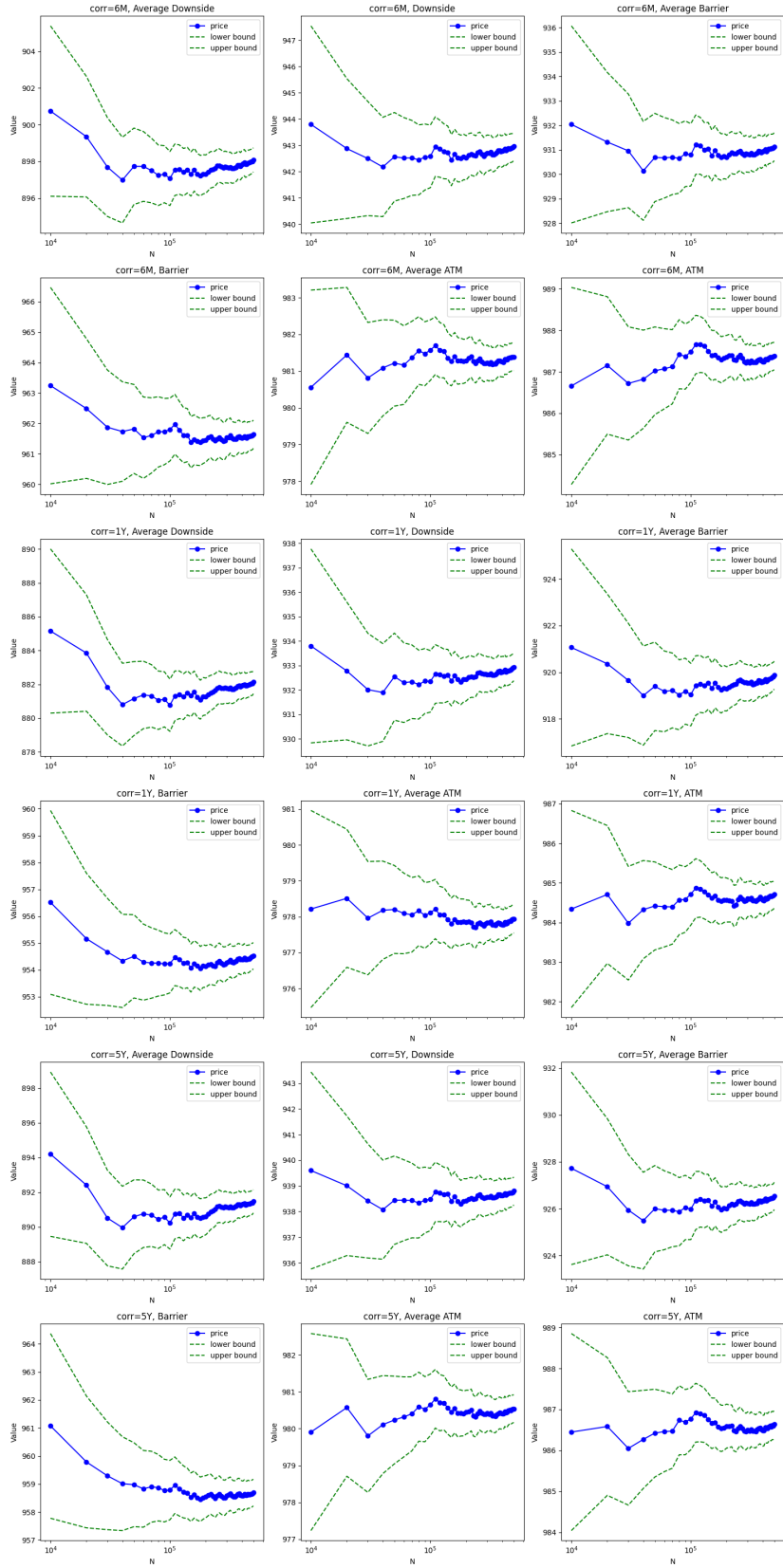


Figure 1: Sensitivity to Number of Simulations

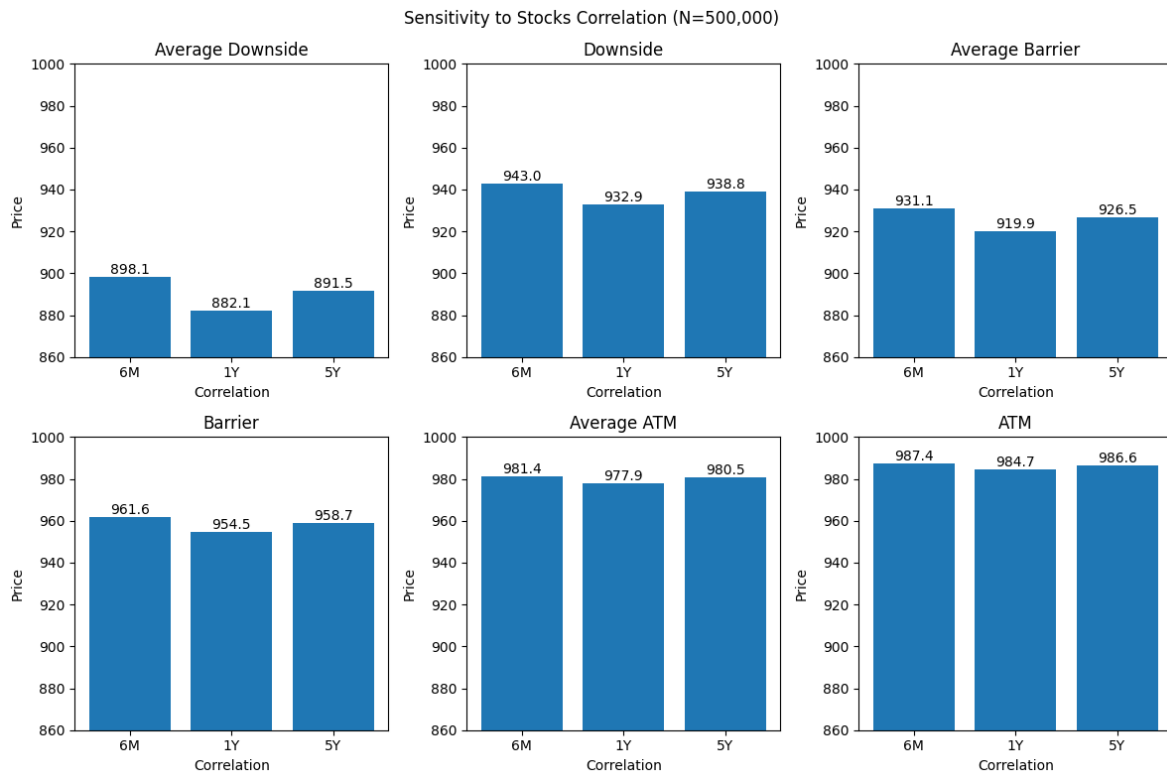


Figure 2: Sensitivity to Correlations between Stocks

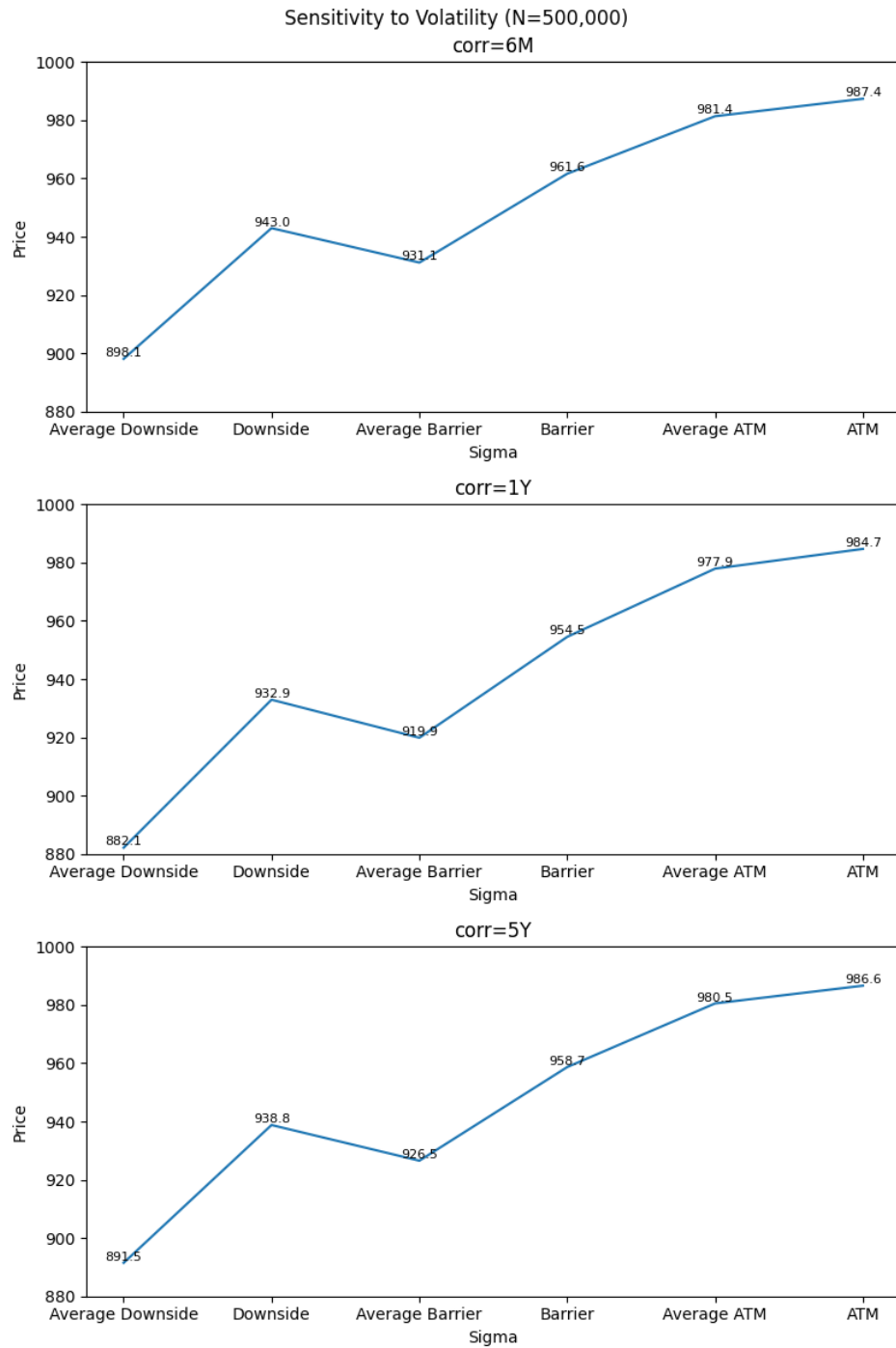


Figure 3: Sensitivity to Volatilities

Appendix: Data from Bloomberg

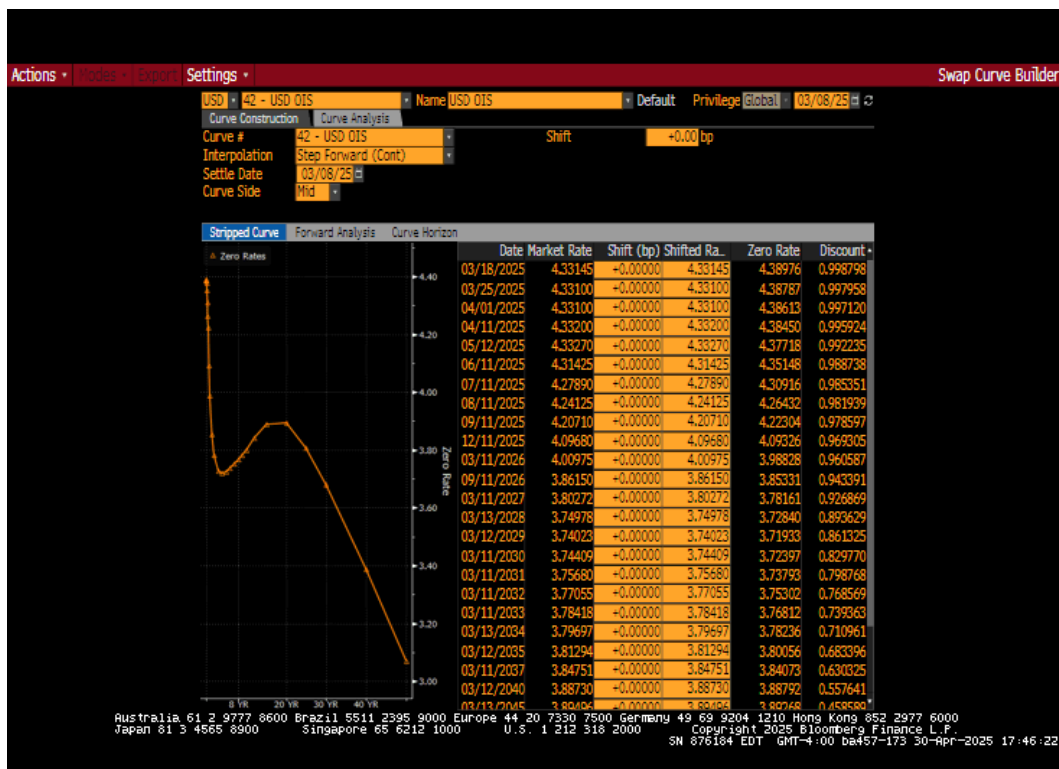


Figure 4: OIS

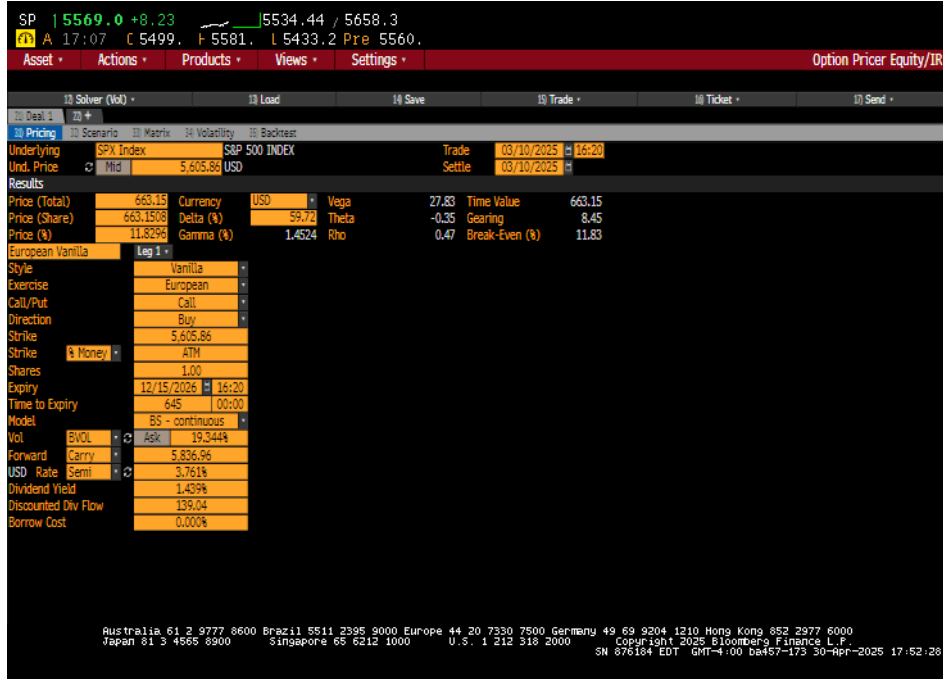


Figure 5: SPX Dividend

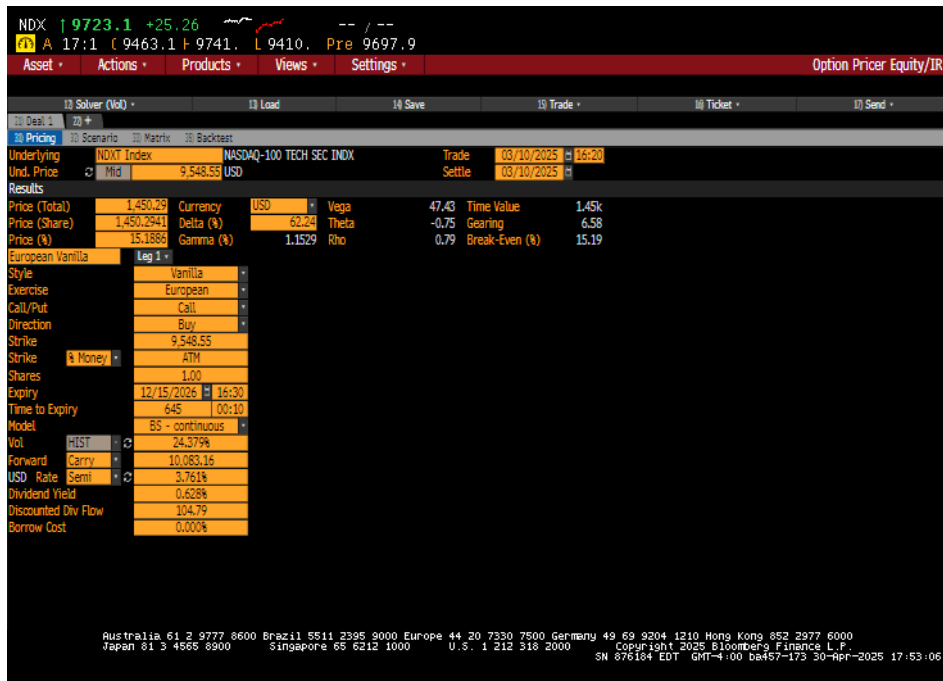


Figure 6: NDXT Dividend

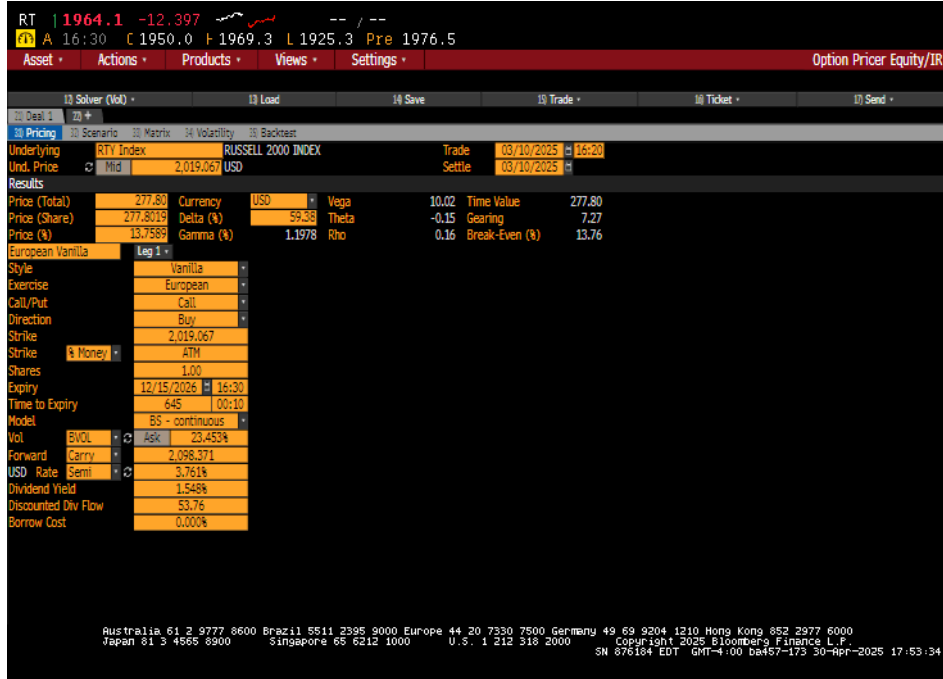


Figure 7: RTY Dividend

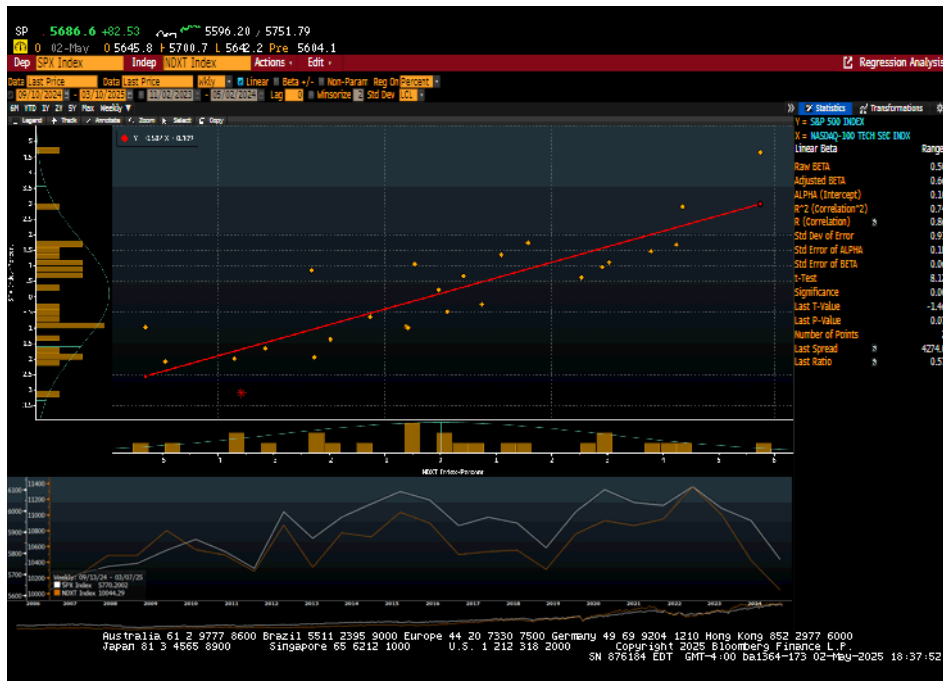


Figure 8: 6 Months Correlation between NDXT and SPX



Figure 9: 6 Months Correlation between SPX and RTY



Figure 10: 6 Months Correlation between NDXT and RTY





Figure 13: 1 Year Correlation between NDXT and RTY



Figure 14: 5 Years Correlation between NDXT and SPX

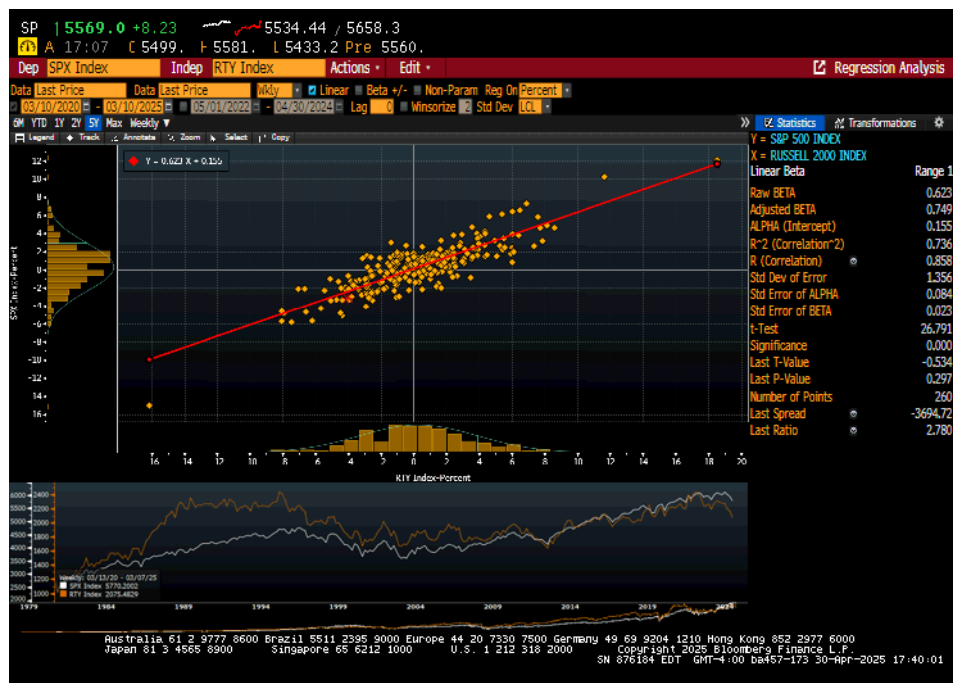


Figure 15: 5 Years Correlation between SPX and RTY

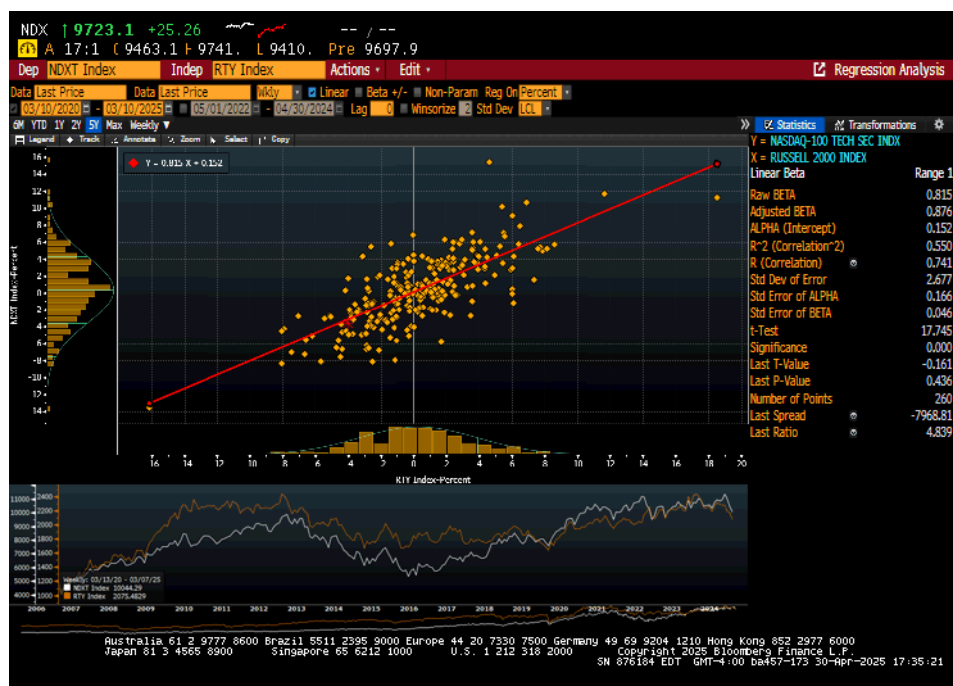


Figure 16: 5 Years Correlation between NDXT and RTY

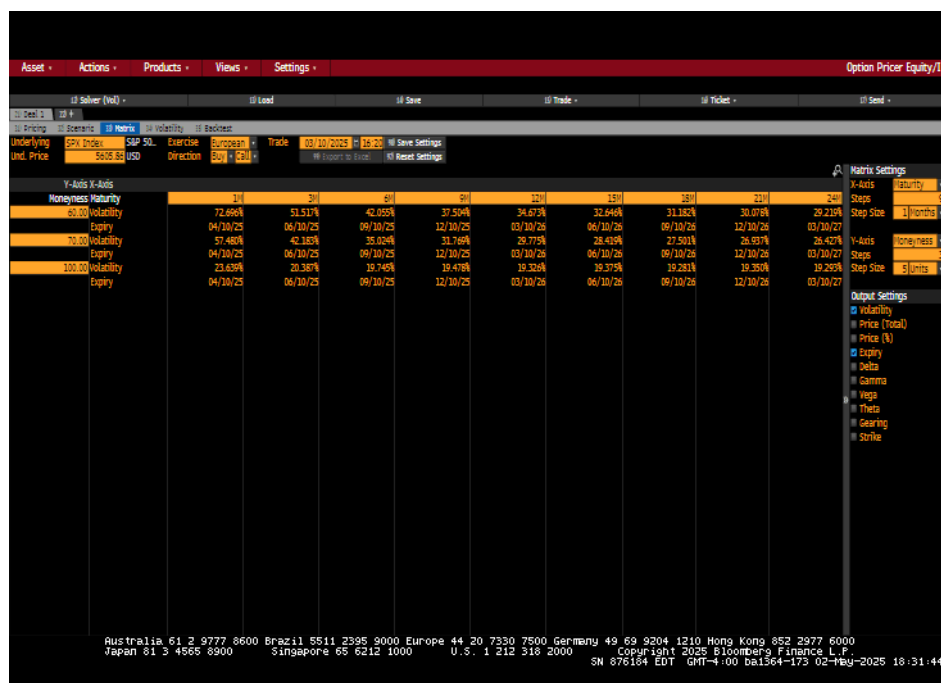


Figure 17: SPX Volatility Surface

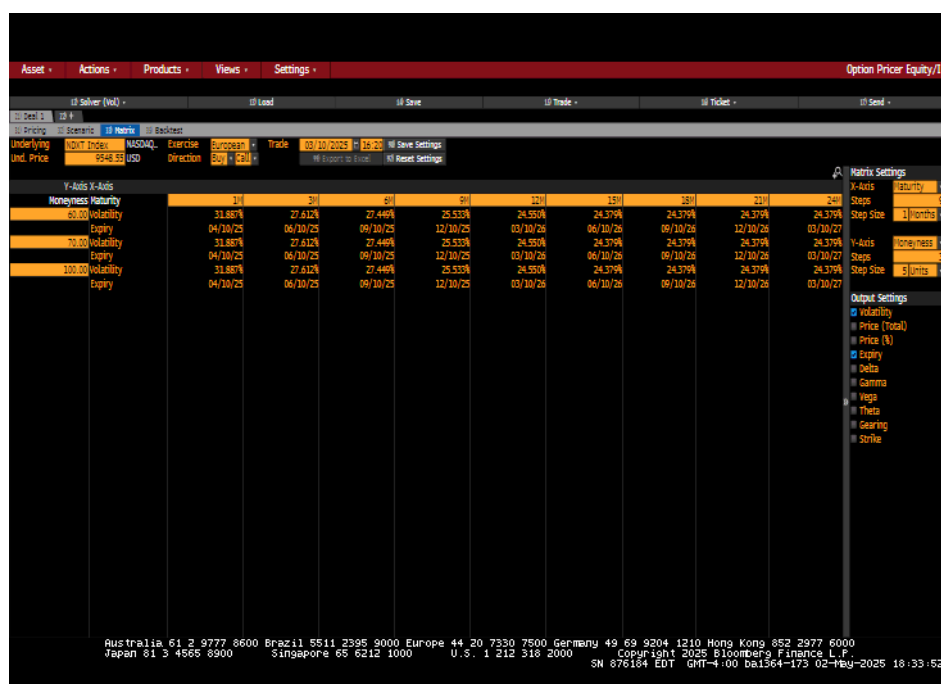


Figure 18: NDXT Volatility Surface

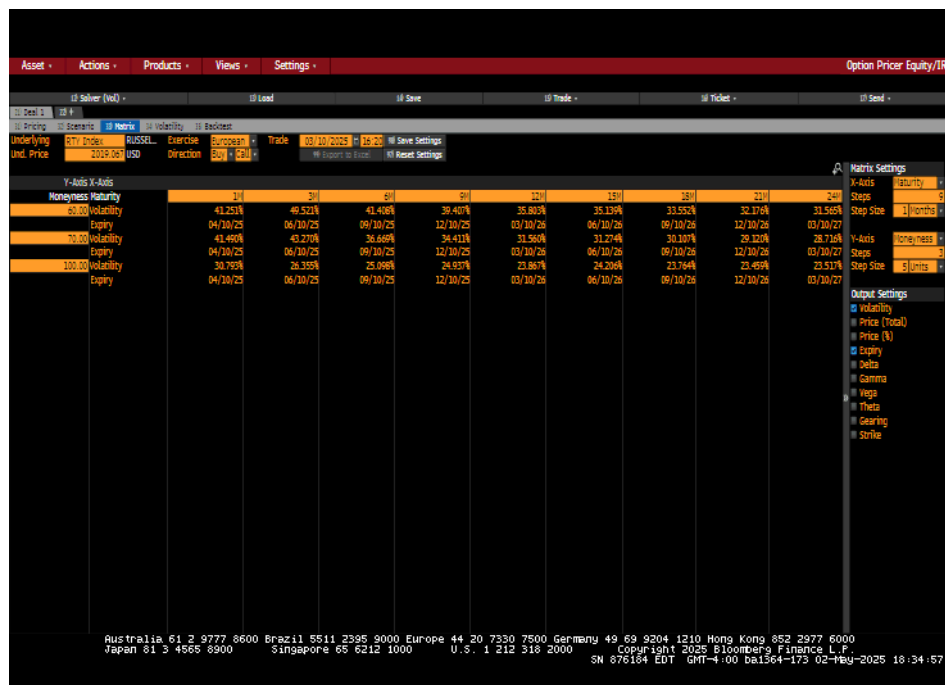


Figure 19: RTY Volatility Surface